Karnajeet Gosavi

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Education

Vishwakarma Institute of Technology, Pune — *Pune, Maharashtra*
Bachelor of Technology in Computer Science Engineering, ***CGPA: 8.98*** — **Jul 2023 – Jun 2027**

Skills

Programming: Python, TypeScript, JavaScript, Java, C++
Frontend: React.js, Next.js, Flutter
Backend: Node.js, Django, Spring Boot, REST API Development
Databases: PostgreSQL, MySQL, MongoDB, Redis
Cloud / DevOps: AWS (EC2, S3, Lambda), GCP, Docker Compose

Experience

Syngenta — **Apr 2026 – Present**
Software Development Intern — *Pune, Maharashtra*

- Automated a SAP CPI documentation generator that converts **7 artifact groups** into one structured DOCX output, including iFlow packages, BPMN/XML flows, schemas, mappings, adapters, Groovy scripts, and external parameters.
- Classified **5 integration layers** before generation to improve traceability across main flows, local subprocesses, message routes, adapter metadata, and error-handling paths.
- Orchestrated a **4-step workflow** for artifact discovery, parsing, AI summarization, and document formatting while supporting up to **80 text artifacts** per package.

Open Source Contributor — **Sep 2026 – Present**
Kornia · OpenMS · jj (Jujutsu VCS) — *Remote*

- Contributed **60+ merged pull requests** across **Kornia** (PyTorch computer vision), **OpenMS** (C++ mass spectrometry), and **jj** (Rust-based VCS), spanning **bug fixes**, **numerical stability patches**, and **new API features**.
- Hardened **numerical precision** by fixing **NaN gradients** in half-precision loss functions and enforcing **float32** computation in **MS-SSIM**; improved error propagation in **jj-vcs**, with additional PRs under active review across all three repositories.

Amazon ML School 2026 — **Jul 1 – Aug 1, 2026**
Participant

- Completed an **8-module curriculum** spanning supervised and unsupervised learning, deep neural networks, reinforcement learning, and generative AI taught live by **Amazon Scientists**.

Projects

InvoX Invoice Financing Platform
Next.js, React, TypeScript, FastAPI, SQLAlchemy, PostgreSQL, Razorpay, Docker

- Architected a full-stack invoice financing marketplace across **21 Next.js pages**, **22 FastAPI route modules**, and **26 SQLAlchemy models** for vendor, lender, admin, invoice, marketplace, chat, and repayment workflows.
- Secured **JWT authentication**, protected routes, vendor onboarding, KYC/document uploads, invoice creation, marketplace listing, lender funding, notifications, and admin actions across role-based workflows.
- Integrated **Razorpay** order creation/verification, invoice PDF generation, smart-contract PDF downloads, blockchain-style invoice registry, and audit trails for funding and repayment transactions.
- Orchestrated **3 ingestion channels** – OCR, Gmail, and Telegram – plus Docker/GCP deployment scripts to support multi-channel invoice intake and cloud-ready deployment.

Whistlechain
Python, TypeScript, FastAPI, React, Algorand SDK

- Built a decentralized whistleblower protection platform with **44 FastAPI routes** across wallet creation, evidence submission, stake tracking, verification, resolution, audit, bounty, and publication workflows.
- Structured **12 backend services** for Algorand testnet access, anonymous wallets, encrypted evidence bundles, IPFS upload, stake management, verification, resolution, audit trail, and bounty payouts.
- Implemented a **964-line PyTEAL evidence registry** with box-based audit records plus commit-reveal verification flows for tamper-resistant complaint lifecycle tracking.
- Shipped a **React 19/Vite frontend** with **19 source files** covering submission, dashboard, inspector portal, admin portal, resolution, audit trail, and wallet connection views.

Anti-Fingerprint Instruction Shuffler
C++, Compiler Backend, Register Renaming, Security

- Created a **C++17 compiler-backend style pipeline** across **54 files**, including 18 C++ sources, 13 headers, and 11 IR samples for code diversification experiments.
- Transformed three-address IR with dependency-safe **instruction shuffling**, CFG-aware **block reordering**, branch fixups, and consistent register/temporary renaming.
- Validated semantic equivalence through runtime verification and a comprehensive harness reporting **PASS=22, FAIL=0, WARN=0** across transformed code variants.
- Generated **fingerprint metrics**, variant ranking, Markdown reports, and HTML reports to compare structural similarity after instruction layout and symbol changes.

Notable Achievements

- Published **3 research papers** in IEEE and Springer venues.
- **Technical and Associate Secretary**, MLSC and Student Development Club 2025; coordinated technical events, workshops, and student development initiatives.
- Won **2 national-level hackathons** across AI, software engineering, and innovation competitions.

Certifications

- IBM DevOps and Software Engineering
- Oracle Cloud Infrastructure 2025 Certified Observability Professional
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional